

Toxic Threat Zone

Time: June 10, 2025 1624 hours PDT (user specified)

Chemical Name: AMMONIA

Wind: 1.55125 meters/second from 271.4° true at 10 meters

THREAT ZONE:

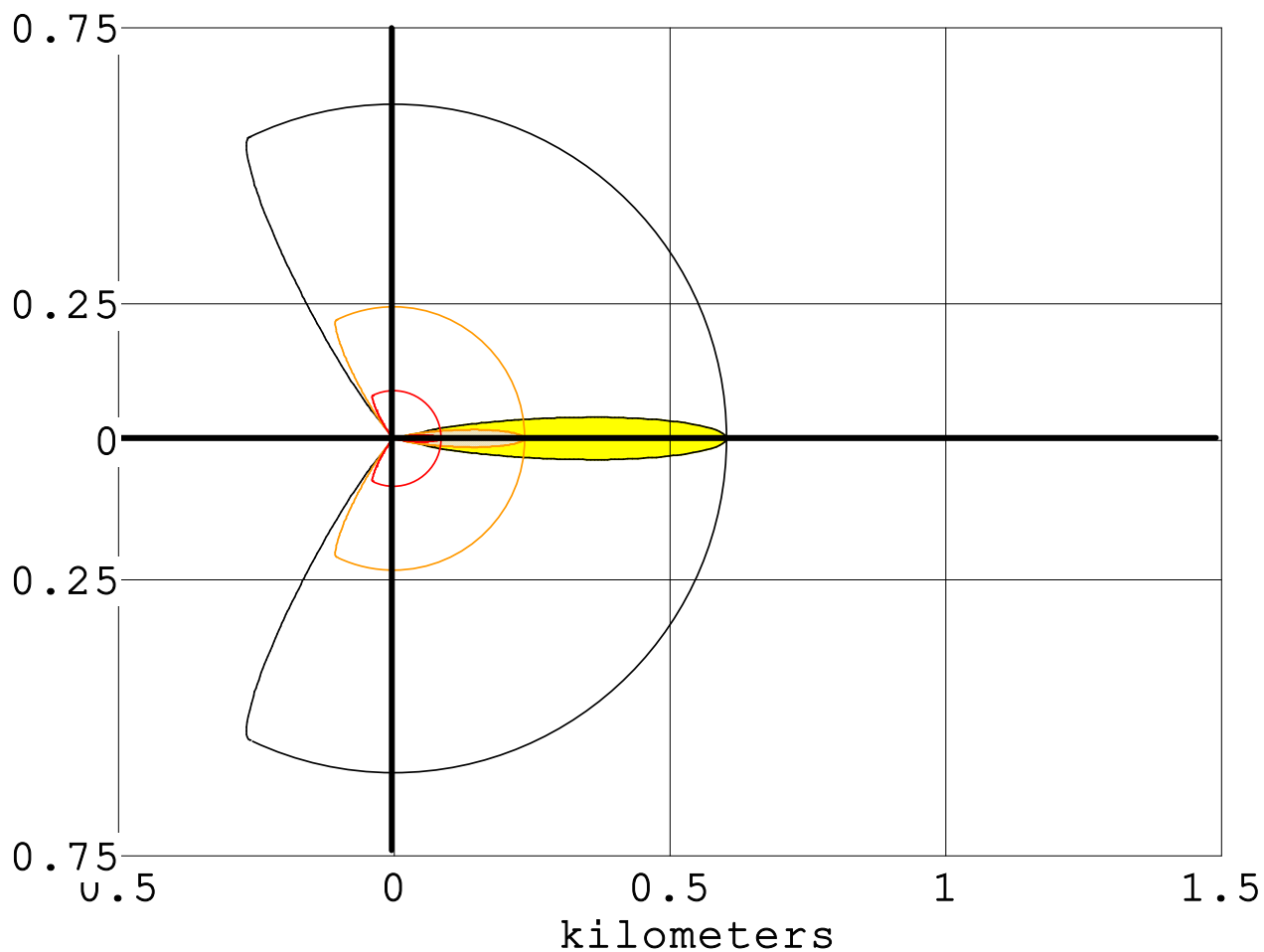
Model Run: Gaussian

Red : 86 meters --- (1100 ppm = AEGL-3 [60 min])

Orange: 238 meters --- (160 ppm = AEGL-2 [60 min])

Yellow: 605 meters --- (30 ppm = AEGL-1 [60 min])

kilometers



-  greater than 1100 ppm (AEGL-3 [60 min])
-  greater than 160 ppm (AEGL-2 [60 min])
-  greater than 30 ppm (AEGL-1 [60 min])
-  wind direction confidence lines

Text Summary

ALOHA® 5.4.7



SITE DATA:

Location: TULARE, CALIFORNIA
Building Air Exchanges Per Hour: 0.28 (unsheltered single storied)
Time: June 10, 2025 1624 hours PDT (user specified)

CHEMICAL DATA:

Chemical Name: AMMONIA
CAS Number: 7664-41-7 Molecular Weight: 17.03 g/mol
AEGL-1 (60 min): 30 ppm AEGL-2 (60 min): 160 ppm AEGL-3 (60 min): 1100 ppm
IDLH: 300 ppm LEL: 150000 ppm UEL: 280000 ppm
Ambient Boiling Point: -33.6° C
Vapor Pressure at Ambient Temperature: greater than 1 atm
Ambient Saturation Concentration: 1,000,000 ppm or 100.0%

ATMOSPHERIC DATA: (MANUAL INPUT OF DATA)

Wind: 1.55125 meters/second from 271.4° true at 10 meters
Ground Roughness: open country Cloud Cover: 4 tenths
Air Temperature: 18.55° C
Stability Class: D (user override)
No Inversion Height Relative Humidity: 54%

SOURCE STRENGTH:

Leak from hole in vertical cylindrical tank
Flammable chemical escaping from tank (not burning)
Tank Diameter: 5.8 meters Tank Length: 12.9 meters
Tank Volume: 341 cubic meters
Tank contains liquid Internal Temperature: 30.6° C
Chemical Mass in Tank: 172,346 kilograms
Tank is 85% full
Circular Opening Diameter: 1 centimeters
Opening is 12.9 meters from tank bottom
Note: RAILCAR predicts a stationary cloud or 'mist pool' will form.
Model Run: traditional ALOHA tank
Release Duration: ALOHA limited the duration to 1 hour
Max Average Sustained Release Rate: 6.13 kilograms/min
(averaged over a minute or more)
Total Amount Released: 368 kilograms
Note: The chemical escaped from the tank as a gas.

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